

	LLM 100	CodeAct 100	RLM 100	LLM 500	CodeAct 500	RLM 500	CodeAct 1000	RLM 1000	RLM Full	
T1.1 Exact product lookup (10Q)	0.99	1.00	1.00	0.97	1.00	1.00	1.00	1.00	1.00	
T2.1 Molecular-weight change (6Q)	0.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	
T2.2 Ring-count change (5Q)	0.02	1.00	0.96	0.02	0.94	0.96	1.00	0.96	1.00	
T2.3 Aromatic-ring formation (5Q)	0.02	1.00	1.00	0.02	1.00	1.00	1.00	0.96	1.00	
T2.4 Combined weight + rings (4Q)	0.00	0.95	1.00	0.00	0.95	0.95	1.00	1.00	1.00	
T3.1 New nitrogen atom (1Q)	0.17	1.00	1.00	0.14	1.00	1.00	1.00	1.00	1.00	
T3.2 C-N formed / C-O broken (1Q)	0.33	1.00	1.00	0.32	1.00	1.00	1.00	1.00	1.00	
T3.3 Single C-C bond formation (1Q)	0.18	0.82	0.84	0.00	1.00	0.37	0.85	0.45	0.55	
T3.4 Fused heterocycle (1Q)	0.25	0.92	0.92	0.17	0.86	1.00	0.97	0.97	0.71	
T3.5 New ring system (1Q)	0.69	0.61	0.49	0.55	0.71	0.60	0.64	0.31	0.32	
T3.6 Fused-ring construction (1Q)	0.00	1.00	1.00	0.00	0.99	1.00	1.00	1.00	1.00	
T3.7 Transition-metal reagent (1Q)	0.80	1.00	1.00	0.54	1.00	1.00	1.00	1.00	1.00	
T3.8 HATU / T3P reagent (1Q)	0.00	1.00	1.00	0.00	0.81	1.00	0.66	0.63	0.95	
T3.9 New stereocenter (1Q)	0.25	1.00	1.00	0.02	1.00	1.00	1.00	1.00	1.00	
T3.10 E-alkene formation (1Q)	0.30	1.00	1.00	0.00	1.00	1.00	1.00	0.99	0.80	
T3.11-14 Amide couplings (4Q)	0.41	0.80	0.73	0.05	0.64	0.65	0.62	0.55	0.41	
T3.15-19 Group transformations (5Q)	0.51	0.92	0.93	0.29	0.86	0.91	0.82	0.82	0.69	
T3.20-21 Protecting groups (2Q)	0.75	0.94	0.97	0.23	0.78	0.79	0.82	0.72	0.85	
T3.22-25 Named reactions (4Q)	0.52	0.86	0.88	0.14	0.85	0.81	0.74	0.69	0.60	
T3.26-30 Mechanisms I (5Q)	0.40	0.78	0.65	0.69	0.87	0.74	0.72	0.63	0.46	
T3.31-35 Mechanisms II (5Q)	0.57	0.73	0.73	0.28	0.61	0.59	0.57	0.56	0.29	
T4.1-2 Fixed-length chains (2Q)	0.00	0.97	1.00	0.00	0.87	1.00	0.93	1.00	0.90	
T4.3-4 Longest chains (2Q)	0.22	0.57	1.00	0.22	0.21	1.00	0.52	1.00	1.00	
T4.5 Hub molecules (1Q)	0.00	1.00	0.80	0.00	1.00	0.80	1.00	0.80	0.98	
T4.6-9 Group-constrained chains (4Q)	0.00	0.75	0.80	0.00	0.90	0.80	0.55	0.75	0.69	
T4.10-11 Protecting-group pairs (2Q)	0.30	0.74	0.90	0.00	0.65	0.91	0.61	0.77	0.56	
T4.12-15 Ring-construction chains (4Q)	0.00	0.62	0.90	0.00	0.51	0.85	0.50	0.95	0.87	
T4.16-25 Truncated routes (10Q)	0.00	0.75	0.66	0.00	0.71	0.60	0.67	0.54	0.00	
T4.26-30 SMIRKS chains I (5Q)	0.27	0.67	0.51	0.00	0.44	0.45	0.34	0.22	0.08	
T4.31-35 SMIRKS chains II (5Q)	0.00	0.76	0.60	0.00	0.44	0.56	0.63	0.46	0.11	